Benchmarking Digitisation Errors in Scalar Field Theories for Quantum Computing

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Simulations of quantum field theories on quantum computers require the digitisation of continuous fields onto finite qubit systems, introducing systematic errors that must be quantified and controlled. In this report, digitisation errors are benchmarked using a harmonic oscillator Hamiltonian in 0+1 dimensions. Three digitisation schemes are compared: Jordan–Lee–Preskill (JLP), its finite difference corrected (FDC) scheme, and a harmonic oscillator (HO) basis truncation. For both free and interacting systems, ground state energies obtained from each scheme are evaluated against known continuum results to quantify digitisation errors. In the free theory, the JLP method significantly outperforms the finite difference approach at fixed qubit number, with discrepancies growing by several orders of magnitude as the number of qubits increases, up to the maximum tested. In the interacting case, truncation in the HO basis yields smaller minimum errors than JLP by approximately one order of magnitude across all qubit numbers considered. These results establish a controlled baseline for assessing whether Hamiltonian Truncation Effective Theory can further reduce truncation induced errors without increasing the qubit budget.

CONTENTS

1. INTRODUCTION

Simulations of quantum field theories on quantum computers are constrained by the digitisation and truncation required to map continuous fields onto finite qubit systems, introducing systematic errors that must be controlled. Hamiltonian Truncation Effective Theory (HTET) [1] provides a non perturbative framework for computing quantum field theory observables, but suffers from exponential growth in computational complexity when implemented classically. It is anticipated that quantum computers may eventually overcome this limitation.

This work represents an intermediate step towards this goal by assessing whether HTET based approaches can lead to more accurate digitisation with fewer qubits. In this report, we systematically benchmark the Jordan-Lee-Preskill (JLP), finite difference corrected (FDC), and harmonic oscillator (HO) basis digitisation schemes at fixed qubit number for a 0+1 dimensional scalar field theory. This establishes a controlled baseline against which the effect of HTET corrections on truncation errors can be meaningfully evaluated without increasing qubit number.

2. BACKGROUND

This section outlines the theoretical framework used to benchmark digitisation errors, introducing the 0+1 dimension scalar field model, the digitisation schemes considered, and the role of Hamiltonian Truncation Effective Theory within future work.

2.1. Scalar Field Theory in 0+1D as a Benchmark

Digitisation errors are benchmarked using a real scalar field theory in 0+1 dimensions, for which the Hamiltonian takes the form of a quantum harmonic oscillator with a quartic self interaction term:

$$\bar{H} = \frac{1}{2}\bar{\Pi}^2 + \frac{1}{2}\bar{\phi}^2 + \frac{\bar{\lambda}_0}{4!}\bar{\phi}^4. \tag{1}$$

The absence of spatial degrees of freedom eliminates lattice spacing and finite volume effects, allowing digitisation errors arising from field truncation and discretisation to be isolated. The model therefore provides a controlled theoretical testbed, often used in digitisation studies, in which errors are explicit and can be reduced systematically.

2.2. Digitisation Methods

2.2.1. Jordan-Lee-Preskill (JLP) and Finite Difference Corrected (FDC)

JLP digitisation is a widely used method [2, 3] and is defined by placing a restriction on the domain of the field $[-\phi_{max}, \phi_{max}]$ which then determines a sampling grid of uniform spacing:

$$\delta\phi = \frac{2\phi_{max}}{n_\phi - 1} \tag{2}$$

where $n_\phi = 2^{n_Q}$ is the number of possible states able to be represented by a number of qubits n_Q. Once the field space is defined the conjugate momentum operator is implemented by performing a quantum Fourier transform (QuFoTr) across the field space. An alternative implementation replaces a QuFoTr based momentum operator with a finite difference approximation acting directly in field space. This avoids explicit basis transformations between conjugate representations however introduces additional discretisation errors associated with a finite difference approximation. These errors can be reduced by introducing polynomial corrections to the finite difference conjugate momentum operator.

2.2.2. *Digitisation Optimisations of Jordan-Lee-Preskill*

Digitisation of a continuous scalar field may be viewed as the combination of two effects: field space truncation imposed through a finite range $|\phi| \leq \phi_{max}$ and a finite resolution determined by n_Q used to encode the field. For a harmonic oscillator with the ground state wave function $\psi_0(\phi) \propto e^{-\frac{\phi^2}{2}}$, the probability weight of the truncated region of the field is given by

$$\int_{|\phi|>\phi_{max}} |\psi_0(\phi)|^2 d\phi \sim e^{-\phi_{max}^2} \tag{3}$$

and therefore, for sufficiently large ϕ_{max}, the error introduced by truncation is exponentially suppressed. Conversely, at fixed n_Q increasing ϕ_{max} enlarges the sampling grid spacing $\delta\phi$, leading to a reduction in momentum space resolution. This effect is explicit within the JLP formulation through the construction of the conjugate momentum operator using a QuFoTr, while in the finite difference approach it appears implicitly, where the accuracy of derivatives is limited by $\delta\phi$.

The optimal balance of the field space truncation and momentum space resolution is constrained by the Nyquist-Shannon (NS) sampling theorem which allows for an optimal choice of ϕ_{max} as a function of the field resolution

$$\phi_{max} = \frac{1}{\sqrt{\bar{\omega}}}\sqrt{\frac{\pi(n_\phi - 1)^2}{2n_\phi}} \tag{4}$$

for a harmonic oscillator from [4] with $\delta x = 1$, corresponding to a choice of dimensionless units after reduction to a single degree of freedom in 0+1D. This expression is exact only in the case $\bar{\lambda}_0 = 0$ in (1) and serves only as an approximation in the interacting case.

2.2.3. *Harmonic Oscillator (HO) basis*

An alternative approach to JLP expands the field in a truncated basis of HO eigenstates. The basis is defined as the eigenstates of a reference HO Hamiltonian,

$$H_{HO}(\omega) = \frac{1}{2}\bar{\Pi}^2 + \frac{1}{2}\omega^2\bar{\phi}^2, \tag{5}$$

with a tunable oscillator frequency ω that sets the width of the basis in field space. This choice is motivated by the form of the interacting Hamiltonian in (1): near its minimum, the potential is approximately quadratic, making the HO basis well suited to describing low energy states that are localised around $\bar{\phi} = 0$.

Whereas JLP digitisation resolves field space uniformly over a finite interval, the HO basis concentrates basis states in regions where the low energy wave functions have significant support. In this representation, truncation occurs in energy space rather than through a hard cutoff on the field amplitude, by retaining only the lowest N oscillator eigenstates. The oscillator frequency ω must be chosen appropriately; when tuned to match the typical curvature of the wave function, low energy states can be captured efficiently while contributions from higher energy states are suppressed by the finite basis size. Although the HO basis does not impose a hard field space cutoff, insufficient basis size effectively limits the representable field space support and momentum curvature. With regards to the NS sampling theorem, the HO basis reduces the number of required basis states by matching the localisation properties of the low energy wave functions in both field and conjugate momentum space, rather than enforcing uniform resolution across the entire field domain.

2.3. Hamiltonian Truncation Effective Theory (HTET)

HTET is then complementary to the process of digitisation, it addresses errors arising from the truncation of the basis rather than from field sampling and thus motivates the investigation pursued in this work. Specifically, HTET corrects errors associated with truncation in energy space due to the omission of high energy basis states. Rather than enlarging the basis directly, HTET accounts for the effects of these excluded states by integrating them out perturbatively. This yields an effective Hamiltonian which acts within the retained low energy space [1]. This framework is particularly well suited to the HO basis used where the basis is explicitly limited to the lowest energy eigenstates. HTET therefore provides a systematic means of incorporating the influence of higher energy states without increasing the basis size.

3. METHODOLOGY

3.1. Digitisation Framework and Benchmarking

We perform a controlled numerical benchmarking of several field digitisation schemes, comparing how digitisation errors scale with the number of qubits used to encode the field. In order to achieve this, several established digitisation methods are considered and benchmarked using a harmonic oscillator Hamiltonian (1). These are: the JLP digitisation and its FDC scheme and the HO basis truncation [2, 3, 5]. At the continuum limit these approaches reproduce the same Hamiltonian and low energy spectrum, however, at a finite qubit number n_Q they exhibit markedly different balances between field truncation and discretisation errors. In practice achieving high accuracy within these schemes requires careful optimisation of parameters which is guided by NS sampling arguments relating field space sampling to momentum space resolution. NS sampling within this work is used as a heuristic guideline and not a strict theorem within the interacting case.

The performance of each method is assessed by comparing the minimum relative error in the ground state energy as a function of the number of qubits used to represent the field, for both non interacting and interacting harmonic oscillator systems. This provides a consistent baseline against which any improvement arising from HTET based modifications to the truncated Hamiltonian can be meaningfully evaluated.

3.2. Implementation

Digitisation methods are evaluated by comparing the calculated ground state energies obtained from digitised systems to known continuum results using the interacting harmonic oscillator Hamiltonian (1). The FDC method is applied only in the free case, $\bar{\lambda}_0 = 0$, in order to isolate errors arising from the finite difference approximation to the kinetic term. The HO basis is applied only in the interacting case with $\bar{\lambda}_0 = 32$, as basis choice is trivial within the free limit as the Hamiltonian is diagonalised, this allows then for a meaningful comparison. The JLP digitisation is applied in both the free and interacting cases as a general comparative method. A value of $\bar{\lambda}_0 = 32$ is chosen to match the strongly interacting benchmark used in [3], ensuring comparability with existing results.

For the JLP and FDC digitisation schemes, the tuning parameter is the field space cutoff $\phi_{\max}$, with truncation occurring directly in field space. For the HO basis, the tuning parameter is the oscillator frequency ω_ϕ, with truncation occurring in energy space through the retention of a finite number of basis states. Each method is evaluated for three values of the qubit number n_Q in order to study the scaling of digitisation errors with available quantum resources. Three values of n_Q were assessed to demonstrate error scaling with qubit number whilst avoiding high simulation times. Comparisons between schemes are then made at their respective optimal tuning values by scanning over the relative tuning parameter (ϕ_{max} or ω_ϕ) and identifying the minimum relative ground state energy error.

Energy eigenvalues are obtained by representing the Hamiltonians as finite matrices in the chosen basis and performing exact diagonalisation using classical numerical methods. The results isolate digitisation errors only. The error metric used throughout is the relative error in the ground state energy, defined with respect to known continuum values [3].

4. RESULTS

4.1. Free Jordan-Lee-Preskill and Finite Difference Corrected Digitisation Errors

Within Figure 1 we observe that for both digitisation schemes, the relative error increases at small and large values of ϕ_{max} with a minimum between these points. At large ϕ_{max} both FDC and JLP tend to the same values of increasing error. As n_Q increases, the minimum shifts to larger ϕ_{max} and lower minimum relative error. The JLP digitisation minima are consistent with the ϕ_{max} values predicted by (4). The FDC method exhibits significantly larger minimum

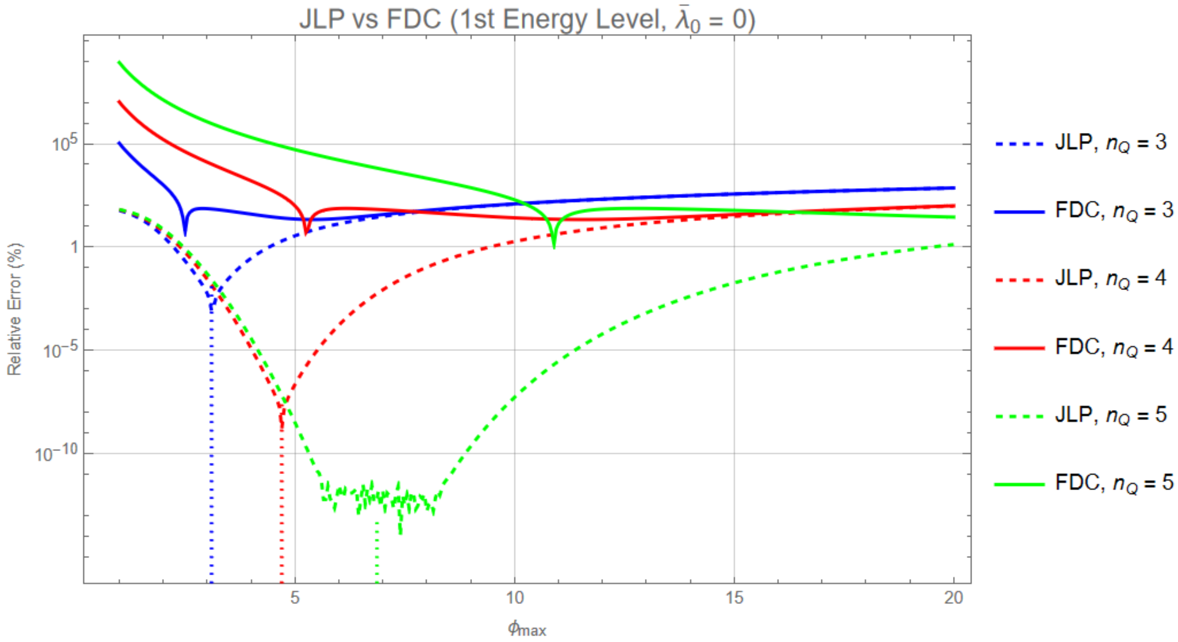

FIG. 1: The expected relative error in the lowest energy state of the harmonic oscillator in (1), with respect to the exact continuum HO energy for an interaction coefficient of $\bar{\lambda}_0 = 0$, on an ideal quantum computer using JLP and FDC digitisation for different values of n_Q as a function of $\phi_{\max}$. The vertical dotted lines indicate the predicted optimal values of $\phi_{\max}$ for JLP for each n_Q given by (4).

relative errors than the JLP digitisation across all values of n_Q considered, with the largest discrepancy occurring at $n_Q = 3$, where the minimum relative error is larger by approximately 11 orders of magnitude. These results isolate only the errors from the finite, truncated qubit representation of the field and conjugate momentum coverage and do not include additional error sources. These results match the qualitative trends produced in [3].

4.2. Interacting Jordan-Lee-Preskill and Harmonic Oscillator Basis Digitisation Errors

Within Figure 2 both digitisation schemes produce a minimum relative error as a function of their respective horizontal axis parameter. An increase in n_Q leads to a decrease in the minimum achievable relative error for both schemes. For all n_Q the HO basis achieves a lower minimum error than JLP by approximately one order of magnitude across each n_Q. In addition, the reduction in the error for each increase in n_Q is more pronounced in the HO basis than in JLP. These results again isolate only the errors from the finite, truncated qubit representation of the field and conjugate momentum coverage and do not include additional error sources. These results match the qualitative trends produced in [3].

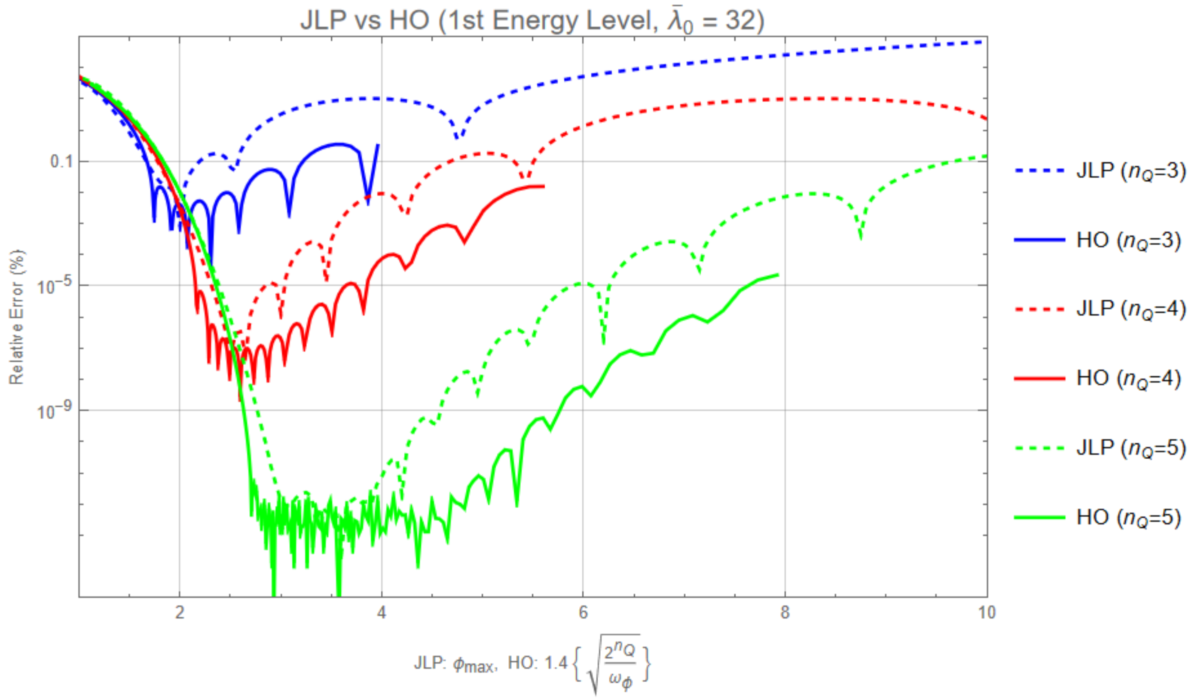

FIG. 2: The expected relative error in the lowest energy state of the HO in (1), with respect to the exact continuum HO energy for an interaction coefficient of $\bar{\lambda}_0 = 32$, on an ideal quantum computer using JLP and HO Basis digitisation for different values of n_Q as a function of ϕ_{max} for JLP and $1.4\sqrt{\frac{2^{n_Q}}{\omega_\phi}}$ for the HO basis.

5. DISCUSSION

5.1. Interpretation of Error Behaviour

In Figure 1, the rapid increase in relative error at small ϕ_{max} is caused by truncation of the wave function, as a significant fraction of the ground state support lies outside the field interval. At large ϕ_{max}, the error again increases, now dominated by discretisation errors from the finite grid spacing $\delta\phi$ at fixed n_Q corresponding to insufficient resolution in momentum space. The minima occur where these two error sources are balanced. For JLP, the predicted optimal values of ϕ_{max} lie at these minima, consistent with (4).

Figure 1 demonstrates that JLP digitisation has significantly lower minimum relative errors than FDC in reproducing the lowest energy level of the harmonic oscillator (1) across all values of n_Q with this disparity becoming larger as n_Q increases. This behaviour can be understood by considering the dominant source of error within each method. In the FDC approach the kinetic term in the Hamiltonian is approximated by a finite difference operator which only becomes exact in the limit $\delta\phi \to 0$. This approximation introduces errors that scale polynomially, leading to a clear saturation in the achievable accuracy even at optimal ϕ_{max}. JLP however uses a quantum Fourier transform which allows the conjugate momentum operator to be represented without a finite difference approximation. Once ϕ_{max} is chosen sufficiently large to contain dominant support for the wave function the remaining digitisation error is dominated by truncation effects which are exponentially suppressed as in (3). This produces the minima seen in the JLP curves and leads to rapidly decreasing minimum errors with increasing n_Q, although with diminishing returns, which is consistent with NS sampling [3]. Therefore for a fixed qubit budget JLP is more efficient than FDC. For JLP however we do note that for $n_Q = 5$ the minimum relative error is not precisely observed as is in [3] due to floating point round off errors at very small relative error values.

Similar to the JLP curves, the HO basis curves in Figure 2 exhibit a minimum in the relative error as a function of the tuning parameter, reflecting a balance between field space localisation and the representation of conjugate momentum for a fixed number of qubits. In contrast to the JLP digitisation, where the field space extent is controlled explicitly by $\phi_{\max}$ and the momentum space resolution is set by the field spacing, the HO basis introduces a correlated dependence between field and momentum space coverage through the oscillator frequency ω_ϕ and the number of retained basis states. Varying ω_ϕ at fixed n_Q therefore trades increased field space localisation against the ability of the truncated basis to represent higher curvature components of the wave function, and vice versa. The minimum in each curve occurs where these competing effects are optimally balanced, with insufficient field space support to the left of the minimum and inadequate representation of kinetic energy contributions to the right. For the values of n_Q shown, an appropriate choice of ω_ϕ allows the HO basis to achieve a smaller minimum relative error than a $\phi_{\max}$ tuned JLP digitisation for this system, consistent with the behaviour reported in [3]. We again note that for $n_Q = 5$ the minimum relative error is not precisely observed as is in [3] for the HO basis due to floating point round off errors at very small relative error values.

5.2. Limitations

This analysis is limited to a 0+1 dimensional harmonic oscillator and considers only digitisation errors in the ground state energy. Errors arising from other sources such as hardware noise and gate level resource costs are neglected, so the reported minima correspond to idealised behaviour rather than realistic quantum computer performance. The HO basis further relies on problem specific tuning that may not generalise to higher dimensional theories. Numerical precision limits also restrict the resolution with which error behaviour can be observed.

6. CONCLUSION

Within this report, the digitisation errors arising from several field digitisation methods used in mapping a continuous scalar field onto a finite qubit system have been benchmarked using a 0+1 dimensional scalar field theory. Digitisation effects have been isolated from other potential sources of error to provide a controlled baseline against which future HTET corrections can be meaningfully assessed. Through comparison of digitisation schemes at fixed qubit number, it is shown that the choice of field space representation has a substantial impact on achievable accuracy even for a simple harmonic oscillator Hamiltonian. In free theory Jordan-Lee-Preskill (JLP) digitisation significantly outperforms its finite difference approach, with a momentum operator implemented via a quantum Fourier transform rather than a finite difference approximation, JLP digitisation errors are exponentially suppressed once NS sampling is saturated for the states of interest; with the finite difference corrected (FDC) kinetic term they remain polynomial in $\delta\phi$. In the interacting case, the Harmonic Oscillator (HO) basis yields smaller minimum relative errors than JLP for the same number of qubits, reflecting the improved efficiency of the energy space truncation for systems with well localised low energy states.

For the systems considered these results demonstrate that digitisation errors are not solely determined by qubit count, but depend critically on how field and conjugate momentum support are distributed. In particular, the enhanced performance of the HO basis at fixed n_Q motivates its use in conjunction with HTET, where corrections act naturally within energy space. The benchmarks presented here therefore provide a validated starting point for assessing whether HTET can systematically reduce truncation errors without increasing qubit requirements. This will be the focus of subsequent work.

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